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Nickel tungstate nanoparticles synthesized *via* a surfactant-assisted method: an efficient platform for neurotransmitter detection

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Accurate detection of dopamine (DPM) is critical due to its essential role in neurological function and its occurrence across diverse biological and food matrices. In this work, we report a novel cetyltrimethylammonium bromide (CTAB)-assisted synthesis of nickel tungstate nanoparticles (NiWO₄ NPs) for the electrochemical determination of DPM in food samples such as yogurt and raw chicken. The successful formation of NiWO₄ NPs was confirmed through XRD, FTIR, and TEM analyses. Electrochemical characterization revealed that the NiWO₄ NP-modified GCE exhibited a significantly reduced charge transfer resistance ($R_{ct} = 165.25 \Omega \text{ cm}^2$) compared with the bare GCE, along with an enhanced electrochemically active surface area (0.126 cm² vs. 0.092 cm²) in the [Fe(CN)₆]^{3-/4-} (5 mM)/KCl (0.1 M) redox system. Amperometric *i*-*t* studies demonstrated a broad linear detection range from 0.001 to 719 μM with a very low detection limit of 0.47 nM. Furthermore, the NiWO₄/RDE sensor displayed excellent selectivity, operational stability, reproducibility, and long-term stability. Real-sample analysis confirmed its practical applicability, with DPM recoveries ranging from 95.54 $\pm$ 0.01% to 99.2 $\pm$ 0.37% in yogurt and chicken samples. This work introduces a CTAB-mediated strategy that enhances the biosensing capability of NiWO₄ nanoparticles through improved electrochemical properties, offering a promising and real-sample-compatible platform for effective dopamine sensing.

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Introduction

Dopamine (DPM), a key member of the catecholamine family, plays a vital role as a neurotransmitter in the central nervous system, cardiovascular system, hormonal regulation, and renal function.¹ Chemically known as 3,4-dihydroxyphenylethylamine, DPM is closely associated with critical neuronal processes such as memory, learning, motivation, and cognition.^{2,3} As an intravenous drug, it is clinically used to elevate blood pressure and heart rate.² In chickens, dopamine is also generated in the enteric nervous system (ENS), produced by the inner layer of the gastrointestinal tract (GIT), where it regulates motility and secretory functions. Gut microbiota influence the biosynthesis of neurotransmitters in the GIT through dietary fermentation pathways. Notably, consumption of fermented dairy products has been linked to reduced anxiety levels in human studies, such as among Azorean university students.⁴ The term *psychobiotics* has been introduced to describe probiotics capable of producing neuroactive compounds, including serotonin, norepinephrine, dopamine, and γ -aminobutyric acid (GABA). These compounds act on the brain-gut axis and may exert antidepressant effects. Yogurt consumption, partly

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through its influence on dopamine metabolism, has therefore been associated with reduced anxiety.⁵ A deficiency in dopamine is implicated in severe neurodegenerative disorders such as Parkinson's and Alzheimer's diseases.^{6,7} Reduced dopamine levels are also linked to neurological and psychiatric conditions, including attention deficit hyperactivity disorder, Tourette's syndrome, and schizophrenia.^{6,7} Therefore, precise quantification of dopamine is of paramount importance in both *in vitro* and *in vivo* contexts.^{8,9} Literature evidence also suggests that consumption of certain foods, including dairy and poultry products, can modulate dopamine levels in human serum, underscoring the relevance of real-sample validation in food matrices such as yogurt and chicken.^{10,11}

Over recent decades, a variety of analytical methods have been developed for dopamine detection in biological samples and pharmaceutical formulations. These include enzymatic assays,¹² fluorimetry,¹³ capillary electrophoresis,¹⁴ high-performance liquid chromatography,¹⁵ UV-visible spectrophotometry,¹⁶ and electrochemical approaches.^{17–20} Among these, electrochemical sensing has gained particular attention owing to advantages such as simplicity, cost-effectiveness, rapid response, and high sensitivity.^{17–21} Neurotransmitters are notoriously difficult to electrochemically detect in biological samples owing to factors such as their extremely low physiological concentrations, the existence of competing species, and electrode fouling brought on by complicated biological matrices. Thus, it is crucial to modify glassy carbon electrodes (GCEs) in order to improve sensitivity, selectivity, and electron-transfer reactions. Due to their low cost, large potential window, and chemical stability, GCEs find extensive use. On the other hand, bare GCEs have high overpotentials for dopamine redox processes and aren't very selective since frequent interferents like ascorbic acid and uric acid have overlapping oxidation potentials; this makes it hard to detect DPM.²² These limitations necessitate surface modification strategies to enhance electron transfer kinetics and achieve selective and sensitive detection.

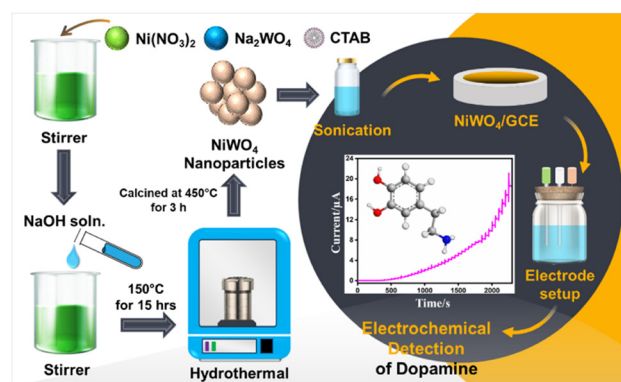
To overcome these challenges, transition metal tungstates are widely explored in electrochemical sensing applications due to their high surface area, tunable electronic properties, and excellent electrocatalytic activity.^{23,24} Within this family, tungstate-based oxides are especially promising, offering structural stability and broad electrochemical utility.^{25–27} Nickel tungstate (NiWO_4), a wolframite-type oxide belonging to the monoclinic $P2_1/c$ space group and sharing structural similarities with $\alpha\text{-WO}_3$, has attracted attention for its small bandgap (~ 2 eV), good electrical conductivity, visible-light responsiveness, and stability.^{28,29} These features render NiWO_4 suitable for diverse applications including photocatalysis, supercapacitors, sensors, and immunosensors.²⁹ Importantly, its strong spin-orbit coupling and unique crystal structure further enhance its potential as an efficient electrode material.³⁰

Surfactant-assisted nanomaterial synthesis has emerged as an attractive strategy. The capacity of cetyltrimethylammonium bromide (CTAB), a cationic surfactant, to adsorb on particular crystal facets during nucleation and growth is one of the

reasons why it is one of the most extensively used surfactants.³¹ This adsorption controls particle aggregation and produces uniform nanostructures.³² CTAB also improves conductivity and electron transmission on electrode materials, which is important for electrochemical applications. It improves sensor sensitivity and reproducibility by stabilizing the nanomaterial surface, enhancing the electroactive surface area, and improving electrode homogeneity. During synthesis, CTAB optimizes the structural and electrochemical properties of nanomaterials easily and effectively.^{33,34}

Several synthesis routes for NiWO_4 nanoparticles have been reported, such as coprecipitation, molten salt, citrate-complex, polymeric precursor, and hydrothermal methods.^{30,35} Hydrothermal synthesis stands out among these methods due to its efficient use of energy, its capacity to create metastable phases, the high crystallinity of the final product, and the wide variety of morphologies it can achieve, from nanoflowers to hierarchical microstructures.^{36–38} The approach also allows for accurate control of reaction conditions, which results in materials with outstanding electrochemical and physicochemical properties. The hydrothermal treatment makes it possible to manage the nucleation and development of NiWO_4 nanoparticles, which ultimately leads to a uniform particle size, a well-defined morphology, and an increase in crystallinity. The surface area and the number of electroactive sites is both increased as a result of these structural modifications, which ultimately leads to an improvement in the electrochemical performance for DPM detection.

In this study, we employed a CTAB-assisted hydrothermal method to synthesize NiWO_4 nanoparticles for dopamine detection (Scheme 1). The synthesized NiWO_4 NPs were systematically characterized by X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), transmission electron microscopy (TEM), electrochemical impedance spectroscopy (EIS), and cyclic voltammetry. The NiWO_4 NP-modified GCE exhibited a significantly lower charge transfer resistance (R_{ct}) than the bare GCE, indicative of accelerated electron transfer and enhanced electrocatalytic activity. The improved conductivity and surface-active sites of NiWO_4 NPs enabled sensitive dopamine detection in real samples, including yogurt and



Scheme 1 Schematic representation of the synthesis of transition NiWO_4 NPs for the electrochemical detection of DPM.

chicken. Using the standard addition method, excellent recovery rates were obtained, demonstrating the practical applicability and commercial potential of this surfactant-mediated tungstate-based electrochemical platform.

Experimental section

Materials and methods

Materials and instruments used for the synthesis and analysis of each characterization are provided in the SI section.

Synthesis of NiWO₄ NPs

NiWO₄ NPs were prepared by a precipitation-assisted hydrothermal method with slight modifications. In a typical synthesis, 0.1 M Na₂WO₄·2H₂O and 0.1 M CTAB (1.09 g) were dissolved in 30 mL of DI water under constant magnetic stirring for 10 min to ensure homogeneity. CTAB acted as a cationic surfactant to regulate nucleation and growth, thereby controlling the morphology and electrochemical properties of the resulting nanoparticles. Subsequently, 0.1 M Ni(NO₃)₂·6H₂O was slowly added dropwise into the above solution under continuous stirring to initiate precipitation. The pH of the reaction mixture was carefully adjusted to alkaline conditions by adding 0.1 M NaOH dropwise until complete precipitation was achieved. Continuous stirring was maintained to ensure uniform mixing and prevent local supersaturation. The resulting mixture was transferred into a 50 mL Teflon-lined stainless-steel autoclave, sealed tightly, and subjected to hydrothermal treatment at 150 °C for 15 h. This process facilitated controlled crystal growth under high-temperature and high-pressure aqueous conditions. After natural cooling to room temperature, the precipitates were collected by centrifugation and washed several times with alternating DI water and ethanol to remove residual ions, unreacted precursors, and surfactant molecules. The washed product was dried in a hot-air oven at 60 °C overnight to obtain a fine powder. Finally, the dried sample was calcined at 450 °C for 3 h in a muffle furnace under ambient air. The calcination step improved crystallinity, removed residual organics (CTAB), and stabilized the NiWO₄ nanoparticles for subsequent characterization and electrochemical studies.

Fabrication of the NiWO₄ NP electrode

For electrode preparation, the as-synthesized NiWO₄ NPs were first dispersed by weighing 2 mg of the powder into 2 mL of ethanol. The suspension was sonicated for 20 min in an ultrasonic bath to achieve a stable and uniform dispersion. The electrode that was used for operation was a glassy carbon electrode (GCE). It was necessary to mechanically polish the GCE surface on a microcloth pad with alumina slurry (0.05 µm, Buehler polishing equipment) prior to modification in order to achieve a surface that appeared like a mirror. The electrode was thereafter washed in a specific order using ethanol and DI water to guarantee a spotless surface, after which it was rinsed with DI water to eliminate any remaining alumina.

Next, 6 µL of the NiWO₄ NPs dispersion was drop-cast onto the polished GCE surface and dried under ambient conditions for 20 min to form a uniform NiWO₄ NP film. The resulting NiWO₄ NP-modified GCE was subsequently used for electrochemical measurements. Bare GCEs, prepared using the same cleaning protocol but without nanomaterial modification, served as controls.

Electrochemical procedures

Electrochemical measurements were performed using a conventional three-electrode configuration. The NiWO₄ NP-modified glassy carbon electrode (GCE) served as the working electrode, a platinum wire was used as the counter electrode, and an Ag/AgCl electrode (in 3 M KCl) functioned as the reference electrode. The electrochemical oxidation of dopamine (DPM) was investigated in the potential range of −0.1 V to +0.7 V (*vs.* Ag/AgCl) at a fixed scan rate of 0.05 V s^{−1}. All experiments were carried out in 0.1 M phosphate buffer (PB, pH 7.0) as the supporting electrolyte. The same electrolyte composition and electrode configuration were employed for subsequent electrochemical sensing studies of DPM.

Real sample preparation

For real-sample analysis, yogurt and raw chicken samples were purchased from a local market in Bangkok, Thailand. The samples were processed immediately upon collection to minimize degradation.

Yogurt sample. An aliquot of yogurt (1 g) was homogenized with 10 mL of phosphate buffer solution (PB, pH 7.0) using a vortex mixer. The mixture was sonicated in an ultrasonic bath for 30 min to facilitate the extraction of dopamine and related biomolecules into the buffer medium. The homogenized sample was centrifuged at 5000 rpm for 10 min to remove insoluble residues. The resulting supernatant was carefully filtered through a Whatman No. 1 filter paper to eliminate fine particulates, and the clear filtrate was collected in a clean beaker. The prepared filtrate was then directly analyzed using the NiWO₄ NP-modified GCE. The standard addition method was applied by spiking known concentrations of dopamine into the yogurt matrix to evaluate recovery efficiency and validate sensor performance under real-sample conditions.

Chicken sample. Approximately 1 g of raw chicken muscle tissue was finely minced and homogenized in 10 mL of phosphate buffer solution (pH 7.0). The suspension was sonicated for 30 min, followed by centrifugation at 5000 rpm for 10 min. The supernatant was filtered through a Whatman No. 1 filter paper, and the filtrate was collected for analysis. Similar to yogurt, the prepared chicken filtrate was directly subjected to electrochemical measurements using the NiWO₄ NP-modified GCE, and the standard addition method was employed to assess recovery and confirm sensor applicability in complex biological matrices.

Results and discussion

X-ray diffraction (XRD) analysis

The crystalline structure and phase purity of the synthesized NiWO₄ NPs were investigated by X-ray diffraction (XRD). The

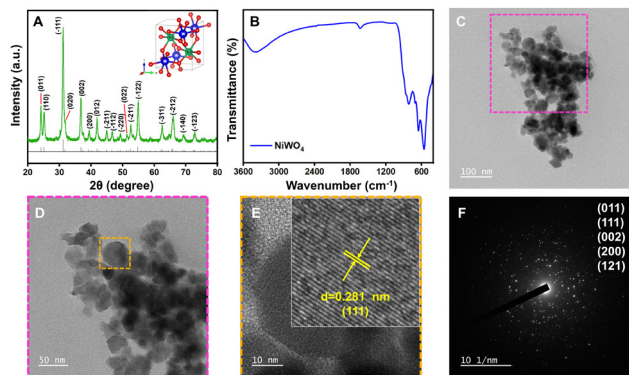


Fig. 1 (A) XRD patterns (inset image shows the crystal structure) and (B) FTIR spectrum of NiWO₄ NPs. (C–E) HR-TEM images and (F) SAED patterns of NiWO₄ NPs.

representative XRD pattern is shown in Fig. 1A. Well-defined diffraction peaks were observed at 2θ values corresponding to the (011), (110), (-111), (020), (002), (200), (012), (-211), (-112), (-220), (022), (-122), (-311), (-212), (-140), and (-123) planes of monoclinic NiWO₄. These reflections are in excellent agreement with the standard Joint Committee on Powder Diffraction Standards (JCPDS) card no. 01-072-0480,^{39,40} thereby confirming the successful formation of phase-pure monoclinic NiWO₄ with a $P2/c$ space group.⁴⁰ The absence of impurity peaks further demonstrates the high purity of the synthesized material.

The sharp and intense nature of the diffraction peaks signifies the high crystallinity of the NiWO₄ NPs and the negligible contribution of amorphous phases. High crystallinity is advantageous in electrochemical applications, as it facilitates efficient charge transfer and minimizes resistive losses. The relative intensity of the (011) and (110) planes suggest preferential crystal growth along these directions, which is consistent with the wolframite-type structure reported for NiWO₄.

The crystallite size was estimated using the Debye-Scherrer equation (eqn (1)).⁴¹

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (1)$$

where D is the average crystallite size, K is the shape factor (taken as 0.9), λ is the X-ray wavelength of Cu K α radiation (1.5406 Å), β is the full width at half maximum (FWHM) of the most intense peak ($\beta = 0.0164$ rad), and θ is the corresponding Bragg diffraction angle. Based on this calculation, the average crystallite size of NiWO₄ NPs was found to be 91.47 nm. This crystallite size is relatively larger than that of some previously reported NiWO₄ nanostructures synthesized *via* hydrothermal or sol-gel methods, where sizes typically ranged between 20 and 60 nm. The comparatively larger crystallite size observed here may be attributed to the longer hydrothermal reaction time (15 h at 150 °C) and subsequent calcination at 450 °C, which promote particle coalescence and grain growth. Nevertheless, the resulting crystallinity and structural integrity of the

NiWO₄ NPs are beneficial for achieving high electrochemical stability during repeated redox cycling.

Fourier transform infrared spectroscopy (FTIR) analysis

The functional groups and bonding characteristics of the synthesized NiWO₄ nanoparticles were analyzed using Fourier transform infrared (FTIR) spectroscopy in the spectral range of 400–3600 cm^{−1} (Fig. 1B) and Table S1.^{42,43} The spectrum exhibits distinct absorption bands that can be assigned to specific vibrational modes of Ni–O, W–O, and O–W–O bonds, which are characteristic of the monoclinic NiWO₄ structure. A prominent band at 548 cm^{−1} is attributed to the stretching vibration of Ni–O bonds in the NiO₆ octahedra, confirming the presence of nickel-centered octahedral units. Bands observed at 869 cm^{−1} and 820 cm^{−1} correspond to W–O stretching vibrations within the WO₆ octahedra, indicating proper incorporation of tungsten in the lattice. Additionally, the bands at 710 cm^{−1} and 639 cm^{−1} are assigned to the asymmetric stretching and bending vibrations of O–W–O linkages in the distorted WO₆ octahedra, forming the wolframite-type framework of NiWO₄. The absence of additional bands corresponding to secondary phases such as NiO or WO₃ indicates high phase purity. These observations confirm the intact NiO₆ and WO₆ octahedral coordination, which is crucial for maintaining the electronic structure and ensuring the high electrochemical activity of the material. Taken together, the FTIR analysis supports the XRD findings, confirming the successful formation of monoclinic NiWO₄ NPs with intact NiO₆ and WO₆ octahedral coordination,^{42,43} which is crucial for maintaining the electronic structure and electrochemical activity of the material.

Morphological analysis

The surface morphology and structural features of the synthesized NiWO₄ NPs were examined by TEM. Representative TEM micrographs captured at different magnifications are presented in Fig. 1C–E. The NiWO₄ NPs predominantly exhibit irregular nano-granular structures, characteristic of particles synthesized *via* hydrothermal methods. The particle size distribution was found to range between ~10 and ~50 nm, confirming nano-scale dimensions and suggesting good surface-to-volume ratios that are favourable for electrochemical reactions.

The high growth density and uniform dispersion of particles indicate successful nucleation and growth under CTAB-assisted hydrothermal conditions. Such a morphology is advantageous in electrochemical sensing because the abundant surface-active sites and short diffusion pathways facilitate rapid electron transfer and improved catalytic efficiency. HRTEM analysis (Fig. 1E) further revealed lattice fringes with an interplanar spacing of 0.281 nm,³⁹ which can be indexed to the (-111) plane of monoclinic NiWO₄, consistent with the XRD results. This correlation between TEM and XRD validates the crystallographic assignment and confirms the high crystallinity of the synthesized material. The selected area electron diffraction (SAED) pattern (Fig. 1F) exhibits sharp diffraction spots arranged in concentric rings, indicative of polycrystalline

NiWO₄ with well-ordered lattice planes of (011), (111), (002), (200), and (121).^{44,45} The spot alignment closely matches the diffraction planes observed in the XRD pattern, further verifying the phase purity and crystalline nature of the material.

Additionally, energy-dispersive X-ray spectroscopy (EDS) elemental mapping (Fig. S1) confirmed the uniform presence of Ni, W, and O across the analyzed region, with no detectable impurities. The homogeneous elemental distribution highlights the compositional purity and structural stability of the NiWO₄ NPs. Collectively, the TEM, HRTEM, SAED, and EDS analyses establish that the synthesized NiWO₄ NPs possess a well-crystallized, nano-granular morphology with uniform elemental distribution, making them highly suitable as an electrode-modifying material for dopamine detection.

Electrochemical impedance spectroscopy (EIS) analysis

Electrochemical impedance spectroscopy (EIS) was employed to investigate the interfacial charge-transfer properties of the bare GCE and the NiWO₄ NP-modified GCE in the presence of 5 mM [Fe(CN)₆]^{3-/4-} containing 0.1 M KCl as the supporting electrolyte. The corresponding Nyquist plots, along with the fitted Randles equivalent circuit, are presented in Fig. 2A. The Nyquist plots consist of a semicircular region at high frequencies and a linear segment at low frequencies. The semicircle diameter corresponds to the charge-transfer resistance (R_{ct}), which reflects the electron-transfer kinetics at the electrode-electrolyte interface. The linear portion represents the Warburg impedance (Z_w), arising from ion diffusion in the electrolyte. The equivalent Randles circuit used for fitting includes the solution resistance (R_s), charge-transfer resistance (R_{ct}), constant phase element (CPE), and Warburg diffusion element (Z_w), providing a comprehensive model for interfacial processes.⁴⁶ For the bare GCE, the R_{ct} value was calculated as

269.9 $\Omega \text{ cm}^2$, indicating sluggish electron-transfer kinetics and poor conductivity due to the lack of active catalytic sites. In contrast, the NiWO₄ NP-modified GCE exhibited a significantly reduced R_{ct} value of 165.25 $\Omega \text{ cm}^2$, confirming its superior electron-transfer ability. The reduction in R_{ct} arises from the high surface area, good crystallinity, and conductive pathways provided by the NiWO₄ nanostructure. The substantial decrease in charge-transfer resistance clearly demonstrates that the NiWO₄ NPs enhance the conductivity and facilitate faster interfacial electron transfer compared with the bare electrode. This improved electrocatalytic activity directly correlates with the enhanced dopamine oxidation performance observed in subsequent electrochemical studies.

Electrochemical behaviors of the NiWO₄ NPs/GCE

To investigate the redox properties of the bare GCE and NiWO₄ NPs/GCE, CV experiments were conducted in the [Fe(CN)₆]^{3-/4-} system (Fig. 2B). The redox responses of the NiWO₄ NPs/GCE and bare GCE were monitored in a 0.1 M KCl solution with 5 mM [Fe(CN)₆]^{3-/4-}. The modified electrodes of the [Fe(CN)₆]^{3-/4-} redox probe device exhibit reversible electron transition, as shown in eqn (2).⁴⁷



The CV findings for NiWO₄ NPs/GCE showed a significant increase in the redox peak current and a decrease in peak-to-peak (ΔE_p) separation than that of the bare GCE. This behavior could be ascribed to the NiWO₄ NPs/GCE electrode's enhanced active surface area, which serves as a feasible electron transport pathway. Additionally, the CV investigation examined the electrokinetic characteristics of the raw GCE and NiWO₄ NPs/GCE. Fig. 2C displays the CV redox current curve of the NiWO₄ NPs/GCE at different scan rates using a redox probe. The scan speeds ranged from 0.02 to 0.2 Vs⁻¹. Besides, the response of the reversible redox peak current *versus* the square root of the scan rate was found with good linearity as shown in Fig. 2D graphs, resulting in linear regression equations with correlation coefficient (R^2) values as shown in eqn (3) and (4).

$$I_{pa} (\mu\text{A}) = 466.64 \left(\frac{V}{s} \right)^{\frac{1}{2}} + 70.951; \quad R^2 = 0.9932 \quad (3)$$

$$I_{pc} (\mu\text{A}) = -331.65 \left(\frac{V}{s} \right)^{\frac{1}{2}} - 61.142; \quad R^2 = 0.9944 \quad (4)$$

The electrochemically active surface area (A) of the NiWO₄ NP-modified electrode is found using I_p in the Randles-Sevcik equation (eqn (5)).⁴⁸

$$I_p = (2.69 \times 10^5) n^{3/2} D^{1/2} A \nu^{1/2} C \quad (5)$$

The number of electrons (n) is provided as 1 in the equation ($n = 1$; for ferrocyanide/ferricyanide), and the diffusion coefficient is denoted as (D). The electrochemical active surface area (EASA) of the NiWO₄ NPs/GCE is 0.126 cm². The larger active surface area affirms the greater electrocatalytic property for the electrochemical sensing DPM.

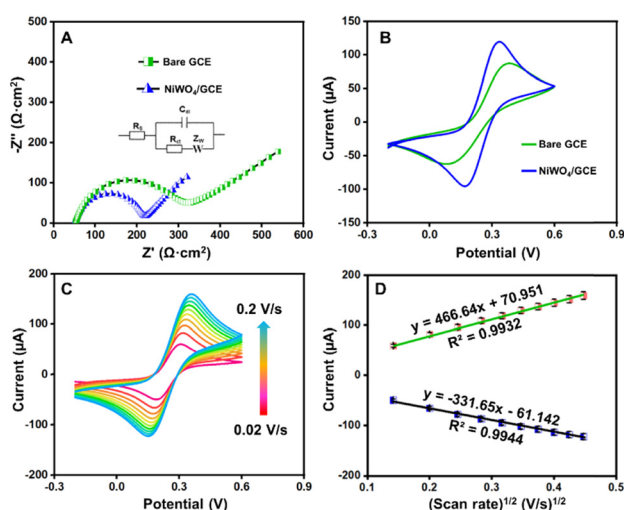


Fig. 2 (A) EIS and (B) CV electrochemical behavior of the bare GCE and NiWO₄ NPs/GCE in the [Fe(CN)₆]^{3-/4-} system. (C) CV of the NiWO₄ modified electrode in the [Fe(CN)₆]^{3-/4-} system at different scan rates (0.02–0.2 V s⁻¹) and the (D) linearity plot of the square root of the scan rate vs. current.

Electrochemical behaviour of the NiWO₄ NPs/GCE towards DPM detection

Electrochemical sensing of DPM on different electrodes.

To determine the suitable working potential for DPM detection, the CV responses of the bare GCE and the NiWO₄ NP-modified GCE were recorded in phosphate buffer (0.1 M, pH 7.0) containing DPM (Fig. 3A). The applied potential range of −0.1 V to +0.7 V was chosen, as it provided a strong current response with minimal background noise, ensuring sensitive detection. Upon addition of DPM, distinct redox peaks were observed at the NiWO₄ NPs/GCE. The prominent anodic peak corresponds to the oxidation of DPM to dopaminoquinone, while the cathodic peak is attributed to the reduction of dopaminoquinone back to DPM. This reversible redox process confirms the electrochemical activity of dopamine on the modified electrode surface. As illustrated in Fig. 3B, the NiWO₄ NPs/GCE exhibited a significantly higher redox peak current compared with the bare GCE. The enhanced oxidation peak current highlights the superior electrocatalytic activity of NiWO₄ NPs, which provide abundant electroactive sites and improved conductivity for dopamine oxidation. Moreover, the sharper and more defined redox peaks at the NiWO₄ NPs/GCE suggest improved reversibility of the redox couple and accelerated electron-transfer kinetics compared to the sluggish response of the bare GCE. These results clearly demonstrate that NiWO₄ NPs impart excellent electrocatalytic properties to the electrode, enabling efficient dopamine oxidation with enhanced sensitivity, reversibility, and electron-transfer kinetics, thereby validating its suitability as a sensing platform for neurotransmitter detection.

Effect of the supporting electrolyte. The pH of the supporting electrolyte plays a crucial role in determining the electrochemical response of dopamine (DPM), as it directly influences both the oxidation potential and current intensity. To study this effect, cyclic voltammetry (CV) experiments were carried out using the NiWO₄ NPs/GCE in 0.1 M phosphate buffer (PB) containing DPM at various pH values ranging from 3 to 11

(Fig. 3C). As shown in Fig. 3D, the oxidation peak current increased progressively as the pH increased from 3 to 7, with a corresponding shift in the peak potential toward less positive values. This enhancement in current indicates more favourable redox kinetics under near-neutral conditions. However, when the pH was further increased from 7 to 11, the oxidation peak current decreased, and the oxidation potential shifted toward more negative values. The reduction in current at alkaline pH may be attributed to excessive deprotonation of the dopamine molecule, which hinders efficient electron transfer at the electrode surface. At higher pH (≥ 9), the prevalence of deprotonated dopamine species alters the electrochemical response, leading to diminished current signals.

From these observations, pH 7 was identified as the optimal condition for dopamine sensing, as it provided the highest oxidation current and well-defined peak response. This pH corresponds to physiological conditions, making the sensor more relevant for real-sample applications. Furthermore, the electrochemical oxidation of DPM was found to proceed *via* a quasi-reversible redox process involving the transfer of two electrons and two protons ($2e^-/2H^+$), as depicted schematically in Fig. 3E. This proton-coupled electron transfer mechanism is consistent with the transformation of dopamine to dopaminoquinone and its subsequent reduction back to dopamine, aligning with earlier reports on dopamine electrochemistry.^{49,50}

Effect of various concentrations and scan rates. Significantly, the anti-fouling properties of the NiWO₄ NPs/GCE were evaluated using different concentrations of DPM (20–100 μ M) at pH-7. Fig. 4A clearly shows that when the concentration of DPM increases, so do the redox peak currents also linearly increase. Fig. 4B shows the calibration plot of the redox peak currents *vs.* various DPM concentrations, which exhibits a linearity with the correlation coefficient as demonstrated in eqn (6) and (7):

Oxidation:

$$I_{pa} (\mu A) = 0.0743 \text{ DPM } [\mu M] + 1.0123; \quad R^2 = 0.9926 \quad (6)$$

Reduction:

$$I_{pc} (\mu A) = 0.0356 \text{ DPM } [\mu M] - 0.4367; \quad R^2 = 0.9945 \quad (7)$$

This proves that the NiWO₄ NPs/GCE possess adequate anti-fouling properties towards different concentrations of DPM. To analyze the impact of the sweep rate, CV analysis was performed using DPM in pH 7 at increasing scan rates ranging from 0.02 to 0.28 $V s^{-1}$ (Fig. 4C). The anodic peak current is observed to thrive linearly with increasing scan rate, and the calibration plot displays the linear equation as shown in eqn (8):

$$I_{pa} (\mu A) = 15.256 \left(\frac{V}{s} \right) + 2.471; \quad R^2 = 0.9956 \quad (8)$$

Furthermore, the cathodic peak current decreases linearly with increasing scan rate, as evidenced by the calibration plot, which exhibits the linear equation as shown in eqn (9):

$$I_{pc} (\mu A) = -15.394 \left(\frac{V}{s} \right) - 0.8327; \quad R^2 = 0.9906 \quad (9)$$

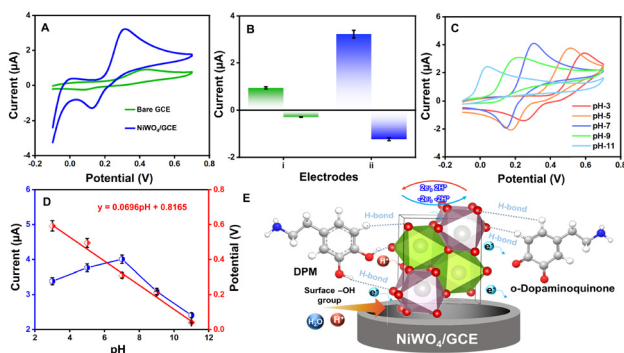


Fig. 3 (A) Electrochemical behavior of the bare GCE and NiWO₄/GCE in the presence of DPM in 0.1 M PB (pH-7) and the (B) bar diagram of the peak current *vs.* modified electrodes. (C) CVs for different pH at the NiWO₄-modified electrode in the presence of DPM and the (D) linearity plot of pH *vs.* peak current and peak potential. (E) Possible electrochemical mechanism of the electro-redox of DPM.

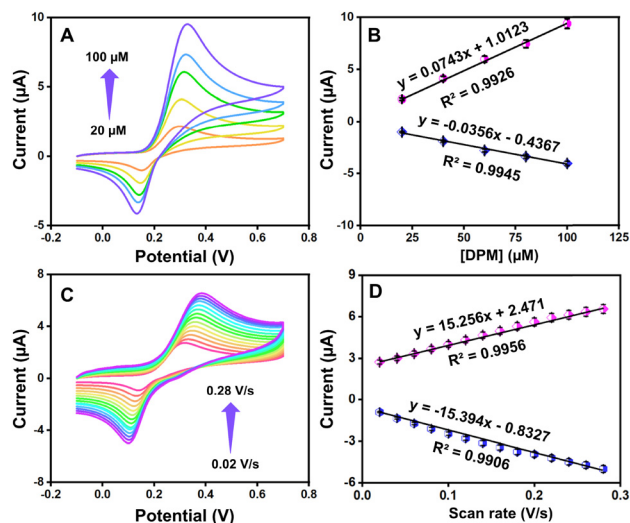


Fig. 4 (A) CVs for different concentrations of DPM (20–100 μM) in pH-7 using the NiWO_4 -modified electrode and the (B) bar diagram of concentration of DPM vs. peak current. (C) Different scan rates (0.02 to 0.28 V s^{-1}) of the NiWO_4 -modified electrode in pH-7 after adding dopamine and the (D) linearity plot of scan rates vs. peak current.

Fig. 4D shows that the scan rate is directly related to the current. Thus, the electro-redox of DPM at the NiWO_4 NPs/GCE surface is controlled by adsorption.

Amperometric i - t studies for DPM detection

Amperometric i - t measurements were performed to evaluate the sensitivity and dynamic response of the NiWO_4 NP/RDE sensor toward DPM. The experiments were conducted by applying a fixed potential of +0.28 V (vs. Ag/AgCl) in 0.1 M phosphate buffer (pH 7.0) while successively adding aliquots of DPM at concentrations ranging from 0.001 to 719 μM . As shown in Fig. 5A, the current response increased sharply and stepwise with each successive addition of DPM, demonstrating the excellent electrocatalytic performance of the NiWO_4 NP/RDE electrode. The inset of Fig. 5A highlights the sensor's sensitivity at low dopamine concentrations, confirming that even nanomolar additions produce well-defined current signals. This rapid and stable current response illustrates the efficient electron-transfer kinetics of the modified electrode.

The corresponding calibration plot (Fig. 5B) demonstrates a strong linear correlation between the oxidation current and DPM concentration over the wide range of 0.001–719 μM . The linearity is described by correlation coefficients of $R^2 = 0.9996$ (low-concentration range) and $R^2 = 0.9945$ (high-concentration range), confirming excellent quantitative performance across both trace and physiologically relevant levels.

The LOD was calculated using the IUPAC criterion (eqn (10)):^{51–53}

$$\text{LOD} = 3 \times \text{standard deviation/slope of the calibration curve} \quad (10)$$

where σ is the standard deviation of the blank and s is the slope of the calibration curve. Based on this calculation, the NiWO_4

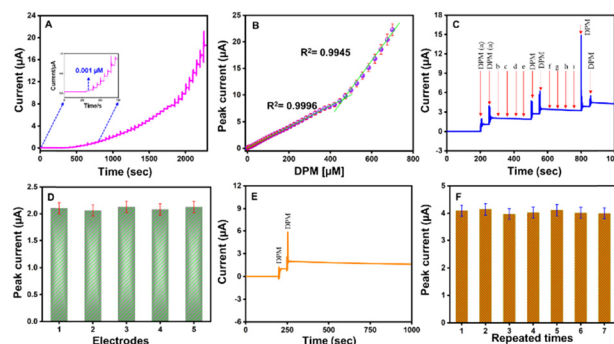


Fig. 5 (A) Amperometric (i - t) measurements of the NiWO_4 -modified electrode after adding DPM (0.001–719 μM) and the (B) linearity plot for the concentration of dopamine vs. peak current. (C) Selectivity study of the NiWO_4 -modified electrode in the presence of DPM and a 5-fold excess concentration of potential interfering compounds. (D) Reproducibility, (E) operational stability, and (F) repeatability studies of the NiWO_4 -modified electrode in the presence of DPM.

Table 1 A comparison analysis for the determination of DPM

Materials	Linear ranges (μM)	LOD (μM)	Ref.
Co,Mo@CNFs	0.01–1000	0.00235	54
ZnNi@f-CNT	200–1200	0.0655	55
ERGO/ImAS	0.1–200	0.03	56
GQDs/IL	0.2–15	0.06	57
$\text{Y}_2\text{O}_3/\text{BN}$	0.05–235	0.006	58
Mo_2C NTs	0.005–50	0.001	59
PT/RGO	0.003–90.206	0.0015	60
MCM/CoWO ₄	10–170	0.0072	61
TCNQ@CoWO ₄	2–80	4.55	62
ZnWO_4/RGO	10–100	4.3	63
La- ZnWO_4	0.10–43.03	0.04	64
CuWO_4	0.6×10^{-8} – 26×10^{-6}	0.024	65
NiWO_4 NPs	0.001–719	0.47	This work

NP/RDE sensor exhibited an ultralow LOD of 0.47 nM for DPM, combined with a wide linear detection range of 0.001–719 μM and the sensitivity of $17.36 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$.

A comparative evaluation with previously reported dopamine sensors (Table 1) indicates that the NiWO_4 NP/RDE sensor outperforms many existing systems in terms of detection limit, sensitivity, and linear range. The superior performance is attributed to the large electroactive surface area of the NiWO_4 NPs, their excellent crystallinity, and the effective electron pathways provided by the RDE. These features facilitate faster ion mobility, enhanced conductivity, and a higher density of catalytically active sites, leading to improved electrochemical response. Overall, the NiWO_4 NP/RDE sensor demonstrates outstanding analytical capability, offering a simple fabrication route, broad dynamic range, and ultrasensitive detection of dopamine. These attributes underscore its potential for the reliable monitoring of neurotransmitters in both clinical and food-related applications.

Selectivity analysis

Selectivity is a crucial parameter in evaluating the reliability of electrochemical sensors, particularly when analyzing complex

biological or food matrices where multiple electroactive species coexist. To assess the selectivity of the NiWO₄ NP-modified RDE toward dopamine (DPM), interference studies were carried out in phosphate buffer (0.1 M, pH 7.0). As shown in Fig. 5C, the current response of the sensor was recorded for DPM in the presence of common potentially interfering substances. A five-fold excess of uric acid, folic acid, ascorbic acid, hydrogen peroxide, glucose, tryptophan, tyrosine, and serotonin were individually added to the electrolyte solution containing DPM. Remarkably, even under these challenging conditions, the NiWO₄ NP/RDE sensor retained a distinct and stable oxidation peak for dopamine, while the interfering species produced negligible current responses. The excellent anti-interference ability of the NiWO₄ NP-modified electrode can be attributed to the strong affinity of DPM molecules for the active surface sites of NiWO₄ and the favourable charge-transfer kinetics at the modified electrode. This selective adsorption and catalytic behaviour effectively suppress overlapping signals from coexisting species, allowing accurate determination of DPM. These findings confirm the high specificity of the NiWO₄ NP/RDE sensor, demonstrating that it can reliably detect dopamine in complex matrices without interference from physiologically or nutritionally relevant electroactive compounds. Such selectivity strongly supports the potential application of the sensor for real-sample analysis of dopamine in biological fluids and food products.

Reproducibility, operational stability, and repeatability studies

The long-term applicability of electrochemical sensors depends on their reproducibility, operational stability, and repeatability. These performance parameters were systematically evaluated for the NiWO₄ NP-modified electrode in DPM detection under physiological pH (0.1 M PB and pH 7.0).

Reproducibility. To evaluate reproducibility, five independently prepared NiWO₄ NP-modified electrodes were tested using CV in the presence of DPM. As shown in Fig. 5D, the oxidation peak currents obtained for each electrode displayed only minor variations, with a relative standard deviation (RSD) of $\pm 4.08\%$. This low RSD value confirms the reliable fabrication process and consistent electrochemical performance of the modified electrodes.

Operational stability. The operational stability of the NiWO₄ NPs was examined using chronoamperometric *i-t* measurements at the optimized potential (Fig. 5E). In the presence of DPM, the electrode exhibited excellent current stability, with only $\pm 9.12\%$ signal loss after continuous operation for 1000 s. This minimal decrease indicates robust electrode-electrolyte interactions and durable catalytic activity under prolonged electrochemical operation.

Repeatability. Repeatability was assessed by performing consecutive CV measurements on a single NiWO₄ NP-modified GCE. As presented in Fig. 5F, the oxidation peak current remained consistent across repeated measurements, with an RSD of $\pm 6.35\%$. This result highlights the excellent electrode stability and reproducible catalytic response during successive cycles.

Taken together, the reproducibility, stability, and repeatability studies demonstrate that the NiWO₄ NP-modified electrode is

Table 2 Recovery ranges of spiked samples

Samples	Added (μM)	Found (μM)	Recovery rate (%)
Chicken	0	—	—
	10	9.554	95.54
	20	19.69	98.45
Yogurt	0	—	—
	10	9.92	99.20
	20	19.792	98.96

highly reliable for DPM detection. Its consistent performance across multiple electrodes, long-term operational stability, and stable repeated responses strongly supports its potential for real-sample analysis and practical electrochemical-sensing applications.

Real-sample analysis

To validate the practical applicability of the developed sensor, the NiWO₄ NP/RDE electrode was employed for the determination of spiked DPM in yogurt and chicken samples.

For sample preparation, 10 mL of homogenized yogurt drink and 10 mL of homogenized chicken extract were transferred into separate 50 mL centrifuge tubes and spiked with known concentrations of DPM. Each sample was vortex-mixed for 10 min to ensure uniform dispersion, followed by storage at -4°C for 1 h to allow stabilization. The samples were subsequently dried under a nitrogen (N₂) atmosphere and re-dissolved in a water/methanol mixture (80 : 20, v/v). To achieve complete homogenization, the tubes were vigorously shaken for an additional 10 min before analysis. To minimize matrix effects and ensure accuracy, the standard addition method was employed for quantification. As summarized in Table 2, the recoveries obtained for both yogurt and chicken samples were consistently high, ranging from 95% to 99%, with low relative standard deviations. These recovery values highlight the reliability of the NiWO₄ NP/RDE sensor in complex food matrices. The excellent agreement between the spiked and measured concentrations confirms that the developed electrode is capable of accurate and selective detection of DPM in real-food samples. This demonstrates its strong potential for practical applications in food quality monitoring and safety assessment.

Conclusions

In this work, NiWO₄ nanoparticles synthesized *via* a CTAB-assisted hydrothermal method were successfully employed for the electrochemical detection of dopamine (DPM). Comprehensive structural and morphological characterization confirmed the formation of highly crystalline monoclinic NiWO₄ with well-defined NiO₆ and WO₆ octahedral coordination and nanoscale morphology. Electrochemical impedance spectroscopy revealed a markedly reduced charge-transfer resistance ($165.25\ \Omega\ \text{cm}^2$) and an enlarged electrochemically active surface area ($0.126\ \text{cm}^2$) at the NiWO₄-modified GCE compared with the bare electrode, confirming its enhanced conductivity and catalytic efficiency. Under optimized conditions, the NiWO₄ NP/RDE sensor exhibited

a broad linear detection range (0.001–719 μM) with an ultralow detection limit of 0.47 nM. In addition to its outstanding sensitivity, the sensor demonstrated excellent selectivity against common interfering species (5-fold excess), robust operational stability, good repeatability, and high reproducibility. Importantly, the NiWO_4 NP/RDE sensor achieved accurate dopamine quantification in real-food matrices, including yogurt and chicken, with recoveries consistently in the acceptable range. Overall, this study establishes NiWO_4 NPs as a promising electrode material for the development of selective, stable, and ultrasensitive electrochemical sensors for dopamine detection. Beyond food analysis, the proposed platform holds considerable potential for applications in biomedical diagnostics and clinical monitoring of neurotransmitters.

Author contributions

Werayut Srituravanich: writing – original draft, supervision, validation, and methodology; Balasubramanian Akila: writing – original draft, visualization, investigation, conceptualization, and data curation; Tse-Wei Chen: supervision, validation, funding acquisition, resources, and methodology; Shen-Ming Chen: supervision, validation, funding acquisition, resources, and methodology; Umesh Narasimha Murthy: writing – review & editing, investigation, and data curation; Elayappan Tamilalagan: investigation, data curation, and formal analysis; Bih-Show Lou: supervision, validation, methodology, funding acquisition, and resources; Sakthivel Kogularasu: writing – review & editing, visualization, and data curation; Guo-Ping Chang-Chien: supervision, validation, and formal analysis.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d5tb02003d>.

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